

Michael S. Albergo

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Address: 99 Stonehedge Lane South, Guilford, CT 06437

EDUCATION

PERIMETER INSTITUTE FOR THEORETICAL PHYSICS

Graduate Fellow

Cambridge, UK

October 2018 - Present

- Applying generative models to improve quantum state tomography. Optimizing to POVM measurement techniques and state fidelity.
- Supervised by Professor Roger Melko, Juan Carrasquilla.

UNIVERSITY OF CAMBRIDGE, DOWNING COLLEGE

MPhil in Physics, focus on machine learning

Cambridge, UK

Expected Graduation Summer 2018

- Main project on implementing generative networks (GANs, VAEs) to simulate detector response for ATLAS detector at CERN.
- Course attendances: quantum computing, quantum information theory
- I supervise the Part III masters course in theoretical particle physics, and I am the welfare officer for Downing graduate students.

HARVARD UNIVERSITY

Bachelor of Arts in Physics: Recommended High Honors, cum laude

Cambridge, MA

May 2017

Secondary in History (focus in 20th Century US)

Further Coursework in Computer Science, Statistics, Data Science, Mathematics, Machine Learning, Healthcare, and Biology

- Weissman Scholar
- Extracurriculars: Head Math Teaching Assistant (Linear Algebra, Multivariable Calculus), Commit 35 hours a week to Hasty Pudding Theatricals Musical, Co-founder of “science and politics” branch of Harvard Institute of Politics

GUILFORD HIGH SCHOOL

Guilford, CT

- Class President, President Spanish Honor Society, graduated with highest honors. ACT 35, Math SAT II 800

June 2013

COMPUTER SKILLS AND PROJECTS

COMPUTER SCIENCE

- **Python, Matlab, C/C++, ROOT**
- Experience with/coursework in supervised and unsupervised learning: generative models, PCA, Bayesian stats, MDP, hidden markov models, neural networks, independent component analysis, canonical correlation analysis
- Maximizing Ebay’s utility for iPhone sales: SVM and logistic regression to optimize sellers’ profit maximizing strategy
- Diagnosing kidney inflammation through convolutional neural network techniques

AGRONOMIA

Co-founder

- Agronomia is an AI-driven marketplace for agricultural waste that biochemical companies can use for feedstock toward creating a more sustainable bioeconomy.
- **1st Place**, European Space Agency “Space for Sustainability” Conference; **1st Place**, Downing College Enterprise Competition

RESEARCH EXPERIENCE

MICROSOFT

Intern, Artificial Intelligence and Research Group

Seattle, WA

June 2018 – September 2018

- Examining non-linear independent component analysis in the context of deep generative models and kernel methods.

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Adjunct Researcher, Cognitive Neuroscience

Cambridge, MA

June 2017 – October 2017

- Dimensionality reduction of semantic space for understanding intrinsic neural sources of language processing in fMRI data.

EUROPEAN CENTER FOR NUCLEAR RESEARCH (CERN)

Researcher, supersymmetry

Geneva, Switzerland

July 2015 – July 2016

- Established kinematic variables to reject QCD background for s-top quark search in 0-Lepton and missing energy final state.
- Created algorithms for data quality of muon detection in the ATLAS experiment for real-time data acquisition

SLAC NATIONAL ACCELERATOR CENTER

SULI Researcher, plasma physics

Palo Alto, CA

June 2016 – August 2016

- Created X-ray propagation and phase retrieval simulation for imaging plasma materials to align theory with experiment.

YALE UNIVERSITY: EMERGENCY MEDICINE

Co-Investigator, medical imaging

New Haven, CT

June 2012 – June 2015

- Co-tailored a convolutional neural network in Matlab to detect the presence and severity of hydronephrosis in ultrasound images to provide automated diagnosis or a validity check on physician decisions.
- Helped determine validity of bedside ultrasound in discerning presence of renal colic to avoid excessive radiation from CT scans.

LEADERSHIP

HARVARD UNIVERSITY MATHEMATICS DEPARTMENT

Head Course Assistant (TA)

Cambridge, MA

September 2014 – May 2017

- Head CA for multivariate calculus class. Teach a weekly section, verify homework solutions, and help designate responsibilities of other CAs. Increased teaching responsibilities for specific students.

HARVARD UNIVERSITY INSTITUTE OF POLITICS

Cambridge, MA

Co-founder of STEAM Initiative

May 2015 – June 2016

- Developed projects with companies and professors to help undergraduates explore the intersection of science and politics.
- Project examples: social media sentiment to beat polling predictions, case study analysis with ADS Ventures on biotech ethics.

MOVING TWICE USA

Cambridge, MA

US Co-Initiator

February 2014 – June 2015

- Moving Twice is an application that brings runners, charities and corporations together in such a way that for every mile a runner using the app runs, a sponsor company will donate a certain amount of money to a given charity.
- Innovated ways to market iPhone app service, acquire corporation sponsors. **Finalist**, Harvard i3 Innovation Competition.